



MOULIE SANKARAN S K

Robotics Hardware Engineer | 3D designing | Prototype Development
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ABOUT ME

Motivated Robotics Hardware Engineer with hands-on internship experience in robotics hardware development, PCB design, embedded systems integration, and AGV/AMR prototyping. Proficient in ROS2-based development on NVIDIA Jetson Orin Nano, ESP32 firmware, multi-sensor fusion, and power management systems. Passionate about building intelligent autonomous systems and contributing to robotics product development in fast-paced engineering environments.

EDUCATION

B.E. — Mechatronics Engineering Kongu Engineering College, Erode, Tamil Nadu CGPA: 7.92	2022 – 2026
HSC Arulneri Thirupani Mandram Higher Secondary School, Erode, Tamil Nadu Percentage: 79.66 %	2021 – 2022
SSLC Arulneri Thirupani Mandram Higher Secondary School, Erode, Tamil Nadu Percentage: 86.88%	2019 – 2020

EXPERIENCE

Design Intern Attox Research Laboratories	May 2024 – May 2025 (12 Months)
- Worked as an intern at Attox Research Laboratories (Attox) from April 2024 to May 2025, contributing to the completion of the Micro+ robotics prototype under the guidance of Director Suganth Murugaraj. - Designed and refined 3D models, mechanical assemblies, and detailed 2D manufacturing drawings for robotics prototypes using SolidWorks and AutoCAD, incorporating design modifications based on prototype testing and review feedback. - During the design internship period, I gained hands-on industrial design experience by working on real-time robotics product development, prototype review cycles, and mechanical design validation processes.	
Hardware Engineering Intern Datacorp Engineering Innovations Pvt. Ltd.	August 2025 – March 2026 (8 Months)
- Worked on robotics hardware prototyping and development for AGV/AMR systems under the guidance of Director Arun Prasath, gaining hands-on experience in robotic system architecture and integration. - Worked with NVIDIA Jetson Orin Nano on Ubuntu 22.04 and configured ROS 2 for sensor communication, hardware interfacing, and motion control applications. - Integrated and tested LiDAR, Intel RealSense cameras, IMU sensors, motor drivers (L298N, BTS7960), OLED displays, and wheel encoder systems, while assisting in hardware bring-up, troubleshooting, and embedded system validation. - During this internship period, I gained practical industrial experience in real-time robotics hardware development, embedded system integration, and prototype testing for AGV/AMR applications.	

TECHNICAL SKILLS

Mechanical Design Tool: SolidWorks, AutoCAD, 3D Designing & Printing, Prusa Slicer, Chitu box
Electrical Design Tool: Proteus, EasyEDA, PCB Fabrication & Designing, Frizting
Embedded Tool: ESP32, Arduino, NVIDIA Jetson Orin Nano, Sensor Integration, Motor Control
Programming: Basics of C Language, Basics of ROS2, Sensor Integration, Basics of Robotics

PROJECTS

IoT-Based Weather Station

2025 | Datacorp, Erode

ESP32, DS80B20, BMP180, AQI, Sound Sensor, Rain Sensor, Wind Speed And Direction.

- Developed a compact IoT-based weather monitoring system using ESP32 with I2C sensor integration for temperature, pressure, air quality, rainfall, sound, and wind analysis, optimized for low power consumption and real-time data display.

DCRA – Educational & Teaching Robot

2025 | Datacorp, Erode

ROS, Mapping & Navigation, Motor Control, Hardware Configuration

- Developed a ROS-based educational robotics platform with autonomous mapping and navigation capabilities, including hardware configuration, sensor integration, and motor control for interactive teaching and learning applications.

Hand Gesture Controlled RC Car (ESP-NOW)

2025 | KEC, Erode

ESP32, ESP-NOW Protocol, L298N, MPU6050

- Developed a gesture-controlled RC car using ESP-NOW peer-to-peer protocol for ultra-low-latency wireless control with MPU6050 tilt-to-drive command mapping.

Hand Gesture Controlled Robotic Arm (ESP-NOW)

2025 | KEC, Erode

ESP32, ESP-NOW Protocol, MPU6050, Servo Motors, Flex Sensor, 3D Print

- Designed a 6-DOF robotic arm controlled by hand gestures using IMU pitch and roll mapping to servo angles through the ESP-NOW peer-to-peer protocol for precise real-time control.

Otto Mini Robot

2024 | KEC, Erode

Arduino, Servo Motors, Ultrasonic Sensor, 3D Printing

- Built a 3D-printed mini biped robot capable of autonomous movement, obstacle avoidance, and programmed dance sequences using servo-driven legs.

CERTIFICATIONS & TRAINING

- CSWA — Completed the Certifications of SolidWorks Associate (International Certification, Dassault Systems- Nov 9 2024)
- In-Plant Training — Cogent Automation Pvt. Ltd., Coimbatore (5 Days, May 2024)
- In-Plant Training — Sri Vela Smelters Pvt. Ltd., Nammakal (10 Days, Dec 2023)

PAPER PRESENTATIONS

- Rapid Peanut Dryer — Automated agricultural dryer with control automation for improved productivity. Anna University Technical Symposium, Coimbatore (Nov 2023).
- E-Bike Design for Sustainable Urban Mobility — Electric bike design for eco-friendly transportation. IES University National Technical Fest, Bhopal (Nov 2024).

LEADERSHIPS & ACTIVITIES

- Design Team (Head) — Robotics Club of Kongu Engineering College (2025–2026): Organized the design team and created innovative content to improve club events and social media attention.
- Joint Treasurer — Robotics Club of Kongu Engineering College (2024–2025): Organised inter-college robotics workshops and technical events
- Active Member — Technical Robotics Society (TRS), KEC (2023–2026)

LANGUAGES

- **English** (spoken & written) **Tamil** (Native)